

HACKATHON

PRODUCT DOCUMENTATION

A complete product, engineering & delivery record

PROJECT IDENTITY

Product / Project Name	[Enter official product name]
Team Name	[Enter team name]
Team Leader	[Full name]
Team Members	[Names + roles]
Mentor	[Mentor name]
Version	1.0
Submission Date	[DD / MM / YYYY]

DOCUMENT CONTROL

VERSION	DATE	AUTHOR / OWNER	CHANGE SUMMARY

02 / EXECUTIVE SUMMARY

The page a product leader should understand in under two minutes.

PRODUCT BRIEF

Product summary — What are you building, for whom, and what measurable value does it create?

DIMENSION	DOCUMENTATION
Problem	
Target user	
Core solution	
Primary value	
Differentiator	

VISION & SUCCESS

Product vision — Describe the desired future state if this product succeeds.

KPI / SUCCESS METRIC	BASELINE	TARGET	MEASUREMENT METHOD

03 / PROBLEM, USERS & OPPORTUNITY

Document the context before explaining the technology.

PROBLEM DEFINITION

Problem statement — State the problem clearly, including who experiences it, when, why it matters, and evidence.

USERS & STAKEHOLDERS

PERSONA / STAKEHOLDER	GOAL	PAIN POINT	SUCCESS LOOKS LIKE
Primary user			
Secondary user			
Admin / Operator			
Other stakeholder			

EXISTING LANDSCAPE

Existing solutions — What do users use today? What gaps, cost, friction or limitations remain?

ALTERNATIVE / COMPETITOR	STRENGTH	LIMITATION	OUR ADVANTAGE

04 / PRODUCT REQUIREMENTS & USE CASES

Translate the problem into a product that can be designed, built and tested.

CORE USE CASES

ID	ACTOR	USE CASE	TRIGGER / INPUT	EXPECTED OUTCOME
UC-01				
UC-02				
UC-03				
UC-04				
UC-05				

FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
FR-01		High	
FR-02		High	
FR-03		Medium	
FR-04		Medium	
FR-05		Low	

NON-FUNCTIONAL REQUIREMENTS

AREA	REQUIREMENT	TARGET / STANDARD
Performance		
Security		

AREA	REQUIREMENT	TARGET / STANDARD
Scalability		
Availability		
Usability / Accessibility		

05 / UX, USER JOURNEY & PRODUCT FLOW

Show how a real user experiences the product from entry to outcome.

USER JOURNEY

STAGE	USER ACTION	SYSTEM RESPONSE	USER OUTCOME
1. Discover			
2. Start			
3. Execute			
4. Receive result			
5. Continue / return			

SCREEN / MODULE	PURPOSE	KEY COMPONENTS	STATE / RULES

END-TO-END WORKFLOW

Insert a workflow diagram or explain the complete sequence: user request → validation → processing → data/AI → response → logging.

06 / SYSTEM ARCHITECTURE & DATA FLOW

This is the engineering blueprint of the product.

ARCHITECTURE DIAGRAM

INSERT FINAL ARCHITECTURE DIAGRAM HERE

Recommended: Client → Frontend → API / Services → Business Logic → Database / AI / External APIs → Monitoring

COMPONENT REGISTER

COMPONENT	RESPONSIBILITY	INPUTS	OUTPUTS	DEPENDENCIES
Frontend				
Backend / API				
Database				
AI / ML				
External Services				
Monitoring / Logs				

Architecture should show trust boundaries, major services, data stores, external integrations and important communication paths.

07 / TECHNOLOGY STACK & ENGINEERING DESIGN

Explain not only what was used, but why it was chosen.

TECHNOLOGY STACK

LAYER	TECHNOLOGY / VERSION	WHY CHOSEN	KEY RESPONSIBILITY
Frontend			
Backend			
Database			
AI / ML			
Cloud / Hosting			
APIs / Integrations			
DevOps / CI-CD			
Testing			

IMPLEMENTATION DECISIONS

Document important design decisions, trade-offs, frameworks, libraries, repository structure and engineering conventions.

CORE LOGIC / ALGORITHMS

Explain algorithms, scoring, recommendations, AI prompts/models, rules or calculations that are central to the product.

08 / DATA, APIS, SECURITY & PRIVACY

Record how information is stored, exchanged and protected.

DATA MODEL

ENTITY / TABLE	KEY FIELDS	RELATIONSHIPS	PURPOSE / RETENTION
User			
Core Entity			
Activity / Transaction			
Analytics / Logs			
Other			

SECURITY & PRIVACY CONTROLS

- Authentication and authorization model; roles and permissions.
- Encryption in transit / at rest and secure secrets management.
- Input validation, rate limiting and protection against common attacks.
- Personal / sensitive data handling, consent and retention policy.
- Third-party access, API keys, audit logs and incident response.

Security risks and mitigations specific to this product.

09 / TESTING, DEPLOYMENT & OPERATIONS

Demonstrate that the product works and can be operated reliably.

TEST STRATEGY

TEST TYPE	SCOPE	TOOL / METHOD	RESULT / EVIDENCE

CRITICAL TEST CASES

ID	SCENARIO	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-01				
TC-02				
TC-03				
TC-04				
TC-05				

10 / IMPACT, ROADMAP & FINAL HANDOVER

Close the document as a product team would: evidence, ownership and next steps.

RESULTS & IMPACT

What was achieved? Include prototype results, user feedback, performance numbers, impact metrics and evidence.

OUTCOME / KPI	RESULT	EVIDENCE / LINK

RISKS, LIMITATIONS & ROADMAP

RISK / LIMITATION		IMPACT	MITIGATION / NEXT STEP
ROADMAP PHASE	TIMEFRAME	KEY ENHANCEMENT	EXPECTED IMPACT
Now			
Next			
Future			

HANDOVER & SUBMISSION CHECKLIST

ITEM	STATUS / LINK
Source repository / GitHub	
Live demo / deployment	
Presentation / pitch deck	
Demo video	
Architecture diagram	
References / acknowledgements	
Mentor approval	

Mentor Sign-off: _____ Date: _____